

Insights Into the Usability of a Digital Art Archive in Mixed Reality

Tiago Martins
Interface Cultures, Institute of Media Design
University of Arts Linz
Linz, Austria
tiago.martins@kunstuni-linz.at

Teresa Romão
NOVA LINCS, DI
NOVA School of Science and Technology / NOVA
University Lisbon
Caparica, Portugal
tir@fct.unl.pt

Abstract

This paper presents preliminary insights into the usability of a digital art archive in mixed reality. By means of a mixed reality headset, archive contents appear in the physical space surrounding the user and can be interacted with using the hands. This virtual archive was created both as an experimental tool and as a companion piece for exhibitions. A small-scale study was conducted to obtain quick insights into the artefact's usability and identify major pain points, which can inform further development of this or other mixed-reality archives. We briefly describe the user experience, as introduced also to study participants, outline the setup of the study carried out in sessions and finally present and discuss results.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**; **Empirical studies in HCI**; • **Information systems** → *Digital libraries and archives*.

Keywords

Mixed Reality, User Experience, Usability, Interaction Design, Digital Archive, Interactive Art

ACM Reference Format:

Tiago Martins and Teresa Romão. 2025. Insights Into the Usability of a Digital Art Archive in Mixed Reality. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3720092>

1 Introduction

The widespread adoption of mixed reality (MR) opens up new possibilities and challenges in the design of user experiences (UX). Users engage with relatively new interaction modes to navigate existing content, such as that made available on the internet. This spatial mode of interacting with content can harness aspects of depth, proximity, haptics and other properties of embodiment in a 3-dimensional space to foster a UX that is more intuitive, efficient and

even fun - also when navigating large collections of heterogeneous media.

In this paper we describe and discuss preliminary usability insights into a digital art archive in mixed reality, manifesting within a physical space and which users can walk through and interact with using their hands. We begin by citing work which relates to our research efforts. We then briefly introduce the context for the design and development of the mixed reality archive, dubbed AR[t]chive, as well as the flow of the user experience. Following that we introduce the goals and setup of the preliminary usability study. Lastly, we present and discuss the outcomes of the study, with considerations towards future developments.

2 Related Work

The near future may come to see wearable displays or 'smart glasses' taking over (or superseding) the role of smartphones as personal tools for communication, information access and entertainment [5]. Presently, in spatial 3D environments such as SteamVR [3], Meta Horizon OS [4] or Apple visionOS [1], legacy content is displayed and navigated essentially as 2D windows in a 3D space.

One may find valuable approaches to visualization and interaction with potentially vast archives within the field of media arts. Medienfluss (or "Media Flow") [6] renders an archive as two parallel flows of media in a large projection, one made up of images and the other of words being spoken aloud via a text-to-speech system. Visitors can interact with the flow by means of a touch-screen console. The project T_Visionarium [7] implements an immersive media space wherein visitors are surrounded by hundreds of video streams of television shows, arranged in a radial swirl-like layout, and which may be sorted and searched according to the emotions they represent. In the works The Living Room and The Living Web [22] immersive CAVE-like environments are employed as intelligent spaces, reacting to visitors' gestures and speech to "engulf" them in images related to their conversations.

In [10] Virtual Research Environments are envisioned as dynamic and ubiquitous environments where researchers can access data and other shared resources from separate institutions, anywhere and at any time. While their vision is mostly based on the web, the introduction of XR technologies opens the space of possibilities, as demonstrated by [11]. Here, an interface design framework for 3D mixed reality environments is presented, based on variations of 2D information spaces, identifying common usage patterns. Similarly, [19] proposes 3D interfaces such as "ring" or "globe" layouts for known-item search in collections of visual data; following with a user study which validates this design, albeit in desktop and mobile

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720092>

devices. [13] investigate opportunities and challenges brought by the use of mixed reality in working environments, such as users being able to create and configure a working space unconstrained by physical limitations. [14] introduce a VR framework which helps users to navigate and understand complex data, adjusting based on their current task. To visualize information, they employ a spherical 3D representation surrounding the user, which their study found valuable. [12] utilize mixed reality to visualize and interact with complex oceanographic data in 3D, using gesture-based controls.

3 User Experience Overview

AR[t]chive was developed as an experimental tool which could also work as a companion piece in exhibitions, within the context of a state-funded partnership project (see Acknowledgements). It is essentially a digital archive in mixed reality, focusing on the works of a specific duo of artists featured on the Archive of Digital Art (ADA, for short) [2]; but it could be developed to include other artists or conceivably adapted to other themes. Its purpose is to inform and entertain, but also to offer a hands-on glimpse into what the experience of exploring and navigating an archive of media can be, in the near future. Our goal was to have a fully functional artefact which visitors could make use of in exhibitions, to further explore the vast oeuvre of the artists beyond the gallery setting, but also to play around with.

The archive runs over a database of contents related to the fifty-plus artworks of artist duo Christa Sommerer and Laurent Mignonneau. It includes images and videos of the works, exhibitions, interviews and even technical sketches and schematics. The database does not include text as such, beyond a succinct description of each work. However, keywords were extracted from related texts, creating additional relations between the works. As part of the overall project, a series of scans of printed media was also produced, in a total of 7900 scans spanning 500 sources (e.g. books, magazines, catalogues) approximately. Additionally, 3D models were also produced portraying several of the artworks which feature in a travelling exhibition.

3.1 Mixed Reality Interface

One key aspect influencing the UX is the interface device itself, the HoloLens 2 developed by Microsoft¹. The display is embedded into a visor which allows the wearer to keep direct visual contact with their surroundings and interposes digital imagery with what the user would normally see. While the wearer retains their entire field-of-view, the actual display area for the “holograms” is perceptibly limited, and virtual objects may appear cut-off for instance when examined up close, or when dealing with large or complex structures.

The main form of interaction with virtual elements is through selection and manipulation using a pinching gesture - by either directly pinching (grabbing) the object or, given the option, using virtual rays which extend from the wearer’s hands. This type of interaction is nowadays quite common in consumer-grade headsets. The HoloLens 2 also periodically scans and keeps track of the physical environment via a Time-of-Flight camera, so that virtual objects may interact with the wearer’s surrounding space.

3.2 UX Flow

We shall now describe the key aspects behind the user experience and interaction design in AR[t]chive, in a manner very close to the guidance given to participants of the study we report further below. This description is kept brief for a matter of scope, both in its original purpose of briefing participants as in the purpose of briefing readers of the present manuscript. A more extensive description, including details and considerations about other aspects of the overall project (such as the production of book scans and 3D models) can be found in a previous publication [18].

When the app starts, the participant is presented with the home scene (Figure 1a), which resembles a galaxy of spheres, each representing an artwork in the archive. Forming a ring at the centre are highlighted artworks, encased in a green bubble. Virtual elements are anchored to the physical space, so the participant is free to move around and reposition themselves as they wish. Work-spheres can be grabbed and pulled apart using both hands. The object will expand into a swirl of media related to that particular work (Figure 1b). All other works in the galaxy will recede and disappear from view.

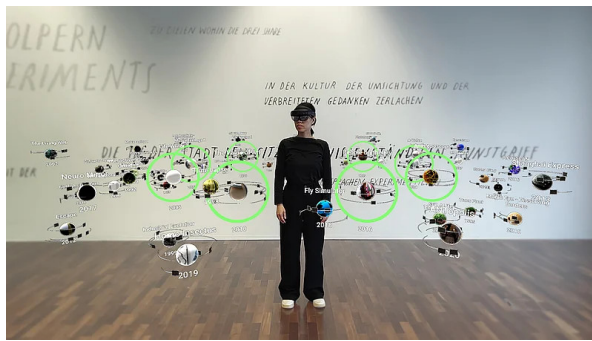
The swirl (Figure 1b) engulfs the visitor in images, videos, keywords, books and also other spheres representing artworks. These can be grabbed, repositioned in space and attached to real surfaces; and some may also be expanded in turn (see below). At the centre of the swirl, one may also find a small descriptive plate with the work’s title, a short summary and (when available) a small 3D model of the installation (Figure 1c). Keywords can also be grabbed with both hands and expanded. As before, other contents recede and disappear, and the keyword expands into a three-dimensional word-cloud of related terms (Figure 1g). Word clouds also contain a few work-spheres for works related to the cloud’s central keyword.

By expanding works and keywords in succession, one may traverse and explore the archive. As a complement, there is also a hand menu which appears when the participant looks at the palm of either hand. Among other items, the hand menu has buttons labelled as “back” and “home” which may be pressed to respectively return the participant to the previous view (swirl, word-cloud or galaxy), or to the initial view (galaxy).

Next to the hand menu there are also a couple of virtual tools. One of these resembles a pin and it is used to create connecting lines between media items to create a mind-map of sorts (Figure 1d). The participant may pick up the pin and bring it close to an item. A dotted line will appear, connecting the pin and the item; and the participant can let go of the pin, which will then attach itself to the item. If the participant picks up a second pin and brings it close to the first one (or any other pin for that matter) a line is created between them.

The other tool appearing next to the hand menu is a small box with a lid, which can be used to store media items (Figure 1e). The participant can grab an item such as an image or keyword with one hand and bring it close to the inventory box on the palm of the other hand. The participant may then let go of the item, which will be stored in the inventory. The participant can pick up the inventory itself and drop it near the ground, in which case the inventory will expand to the size of a small trunk and hover a few feet above the ground (Figure 1f). In this state, items can both be

¹<https://www.microsoft.com/en-us/hololens/>



(a) home or galaxy view



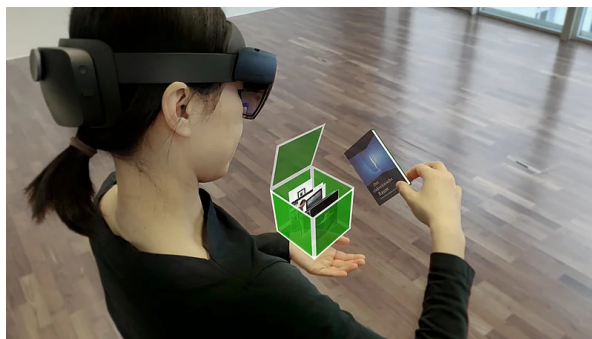
(b) swirl view



(c) swirl view, close-up with 3D model



(d) creating a composition



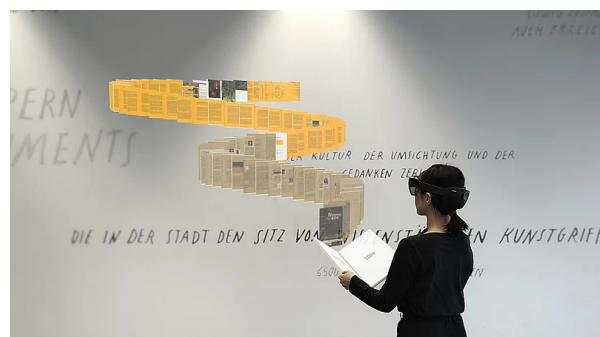
(e) inventory in hand mode



(f) inventory in trunk mode



(g) word-cloud view



(h) book view

Figure 1: core views, tools and interactions in AR[t]chive

added and removed from the inventory. The large inventory can be picked up, in which case it will shrink again, and returned to the palm of the participant's hand.

This description follows the same sequence of the hands-on tutorial given to participants of the user studies (see below). The following however was not explained initially, but it rather offered to participants as they freely explored, as needed, for a matter of brevity and to keep participants focused on the core interactions.

A pin can be removed from the scene by grabbing it and releasing it far from any item (i.e. when no connecting dotted line is visible). This also deletes any line connected to that pin. As such, to remove a connecting line, one of the pins at its end must be disposed of in this manner.

Among the displayed media elements there are also books, which can be opened much like work-spheres and keywords. Books represent sources of printed materials that have been scanned for the archive. As the book opens, scanned pages from that source extend outwards in a waving line, so they may occupy the participant's field of view (Figure 1h). Much like any collection, individual pages can be grabbed and used like any other individual media item.

Swirls, word-clouds and open books are collections of media, and the navigation in AR[t]chive happens from collection to collection. Collections appear anchored in physical space, but they can be moved by grabbing and moving the element at the center: the info plate for a swirl, the central keyword for a word cloud or the book for the open book. Collections can also be closed by grabbing that same central item with both hands and bringing them together, contracting the item as opposed to expanding. This brings the participant to the previous collection (much like the "back" button). The same gesture applied to an individual media item (such as images or keywords) removes it from the scene, which can be useful for tidying up.

4 Usability Assessment

In the following sections we detail the usability study, starting with its goals, and then describing the structure and setting of the study sessions; after which we present and discuss outcomes.

4.1 Objectives

The overall goal of the study was to obtain quick insights into aspects of the artefact's usability and overall experience of users. More concretely, we wished to determine if AR[t]chive is easy to understand and to use; and beyond that if it is also engaging or even fun to use; while rooting out any major pain points. While we did not wish at this stage to evaluate whether AR[t]chive is an efficient tool for research, ideation or mind-mapping (this will require a larger endeavour), we did lay out the main task given to participants in that same context, to encourage the use of most (if not all) possible types of interaction.

4.2 Study Setup

The study was conducted in structured individual sessions, with each participant using AR[t]chive and answering a questionnaire. Sessions lasted approximately thirty-five minutes and took place in a university project room, a space of roughly 6 by 6 meters configured as an unoccupied central space surrounded by desks

with PC workstations, equipment lockers and a few empty walls. In each session two researchers were present: one taking on the role of facilitator and the other the role of observer. The evaluation was accomplished through passive observation of the user interacting with the system, followed by a questionnaire; and a few occasional questions to clarify choices and actions taken by the participant, as needed.

After greeting the participant, answering any pending questions and receiving a signed consent form, the facilitator gave a brief introduction of AR[t]chive, as a digital art archive in mixed reality, and outlined the structure of the session. The participant put on the headset, and was reminded that they could stop the session at any time, without the need for justification. The participant was then given a hands-on introduction to AR[t]chive. The facilitator explained a single interaction at a time and then invited the participant to perform it, in some cases multiple times, very closely to what is described in the UX Flow section above. This introduction lasted around ten minutes.

After the introduction, the participant was given free rein to explore and play around with AR[t]chive for roughly five minutes. They were encouraged to think aloud about their experience and pose any questions they may have. The facilitator also suggested trying out different interactions, such as opening a book.

Following the period of free exploration, the participant was given a concrete albeit open-ended task: to create a composition of materials related to insects (a common motif in the collection of artworks). The composition should have around ten items and each item should be connected to at least one of the others with a line (using the pin tool). The participant was encouraged to use the space, walls and surfaces of the room as they liked, and also to talk aloud about their experience. This task had a duration of ten minutes. The point was not to evaluate the composition per se, but rather to provide a scenario of use of AR[t]chive as a tool with a defined goal in mind.

Finally, after removing the headset, the participant was given a questionnaire to fill out in digital format. The questionnaire was composed first of a few questions focused on the participants' demographic data, as well as on their acquaintance with AR[t]chive, mixed reality and digital technologies; followed by questions from known feedback models (items from the Presence Questionnaire (PQ) [23], the full System Usability Scale (SUS) [9] and the User Experience Questionnaire (UEQ) [16]; and closing with three open-ended questions.

4.3 Results

A total of fourteen volunteers participated in the study, ages ranging from 25 to 60 years old (mean: 33.9, SD: 10.4); with six female, one non-binary and seven male individuals. All participants were closely related to the context of digital media education and research, quite familiar with common digital tools, and all had previously tried out VR or AR experiences for at least a short amount of time. They can be said to represent the core target audience of AR[t]chive as a tool. A majority of participants previously knew about AR[t]chive, and a half of them had also tried it out before the study. However, their knowledge of the system was marginal and all participants received the same hands-on introduction to AR[t]chive, as well as

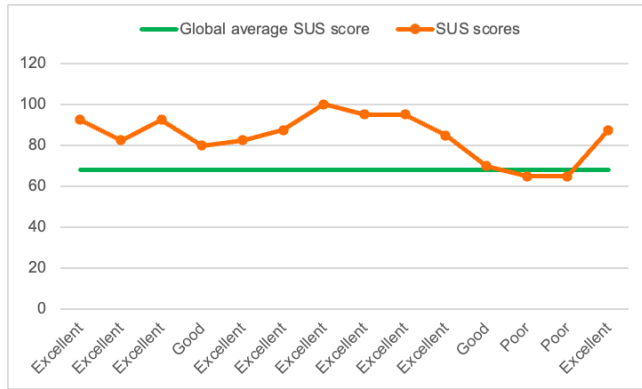


Figure 2: SUS – participants' scores

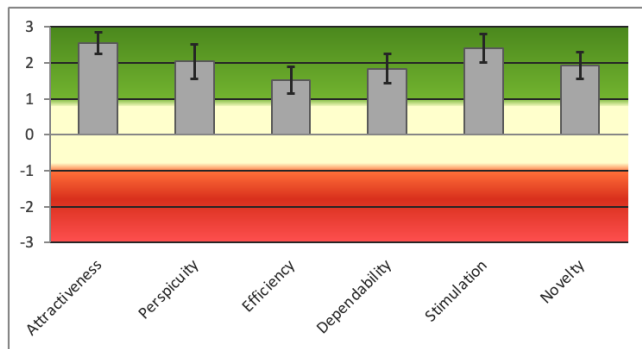


Figure 3: UEQ – scales results

benefitting from a period of free exploration, where they could try out different interactions before being asked to perform specific tasks.

The SUS questionnaire [8, 9] was used to evaluate the overall usability of a system. The resulting SUS score was 84.3, a value much higher than the global average score (68) [17]. More than 70% of participants (10 in 14) considered the usability of the artifact to be excellent (SUS score above 82.5) (Figure 2). These results are in line with the participants' behavior observed during the tests, generally reflecting positive feedback.

The User Experience Questionnaire (UEQ) measures a product's user experience across six dimensions: Attractiveness, Perspicuity, Efficiency, Dependability, Stimulation, and Novelty [21]. The results were very positive (Figures 3 and 4). The dimension with lower average was Efficiency. Even so, AR[t]chive's efficiency was classified as good when compared with the benchmark data set [20]. In the remaining dimensions, most relevant when considering the artefact's goal, AR[t]chive was classified as excellent, with attractiveness displaying the highest score. Thus, AR[t]chive captured the participants' interest, who considered it easy to get familiar with, creative, enjoyable and exciting to use.

An adjusted version of the Presence questionnaire [15] was used for this study. For a matter of time and focus, we selected questions which complement the above two questionnaires, namely those related to Possibility to Act (items 1, 2, 8 and 9) and Possibility to

Examine (items 11, 12 and 19). All items used a 7-point scale, and for all the considered questions "7" was the best result. All questions obtained an average value over 5.5, with the question "How well could you examine objects from multiple viewpoints?" (Q12) being the best rated (mean: 6.29, SD: 0.73); and the question "How closely (i.e. in detail) were you able to examine objects?" (Q11) being the worst rated (mean: 5.5, SD: 1.22).

Participants' responses reveal that they were able to control the events (Q1) (mean: 5.79, SD: 0.7); the environment was responsive to the actions they performed (Q2) (mean: 5.57, SD: 0.94); they were able to anticipate what would happen next in response to the actions they performed (Q8) (mean: 6.07, SD: 1.07); and they were able to actively survey or search the environment using vision (Q9) (mean: 6.21, SD: 0.89). Moreover, participants were able to inspect in detail, the objects available for interaction in AR[t]chive (Q11) (mean: 5.5, SD: 1.22); they could examine objects from multiple viewpoints (Q12) (mean: 6.29, SD: 0.73) and concentrate on the required activities rather than on the mechanisms used to perform them (Q19) (mean: 5.57, SD: 1.34). These results corroborate the observations made by the research team during the tests.

Continued observation allowed us to further analyze the participants' behavior to assess AR[t]chive's positive attributes, as well as aspects to improve. Participants were all very enthusiastic and happy while using AR[t]chive. They displayed different approaches while engaging with it: some were more cautious, structured and thoughtful in their interactions, while others explored in a more fearless and eager manner. This is likely due to their different personality traits. Nevertheless, all seemed very interested, curious and committed to performing their tasks and exploring the system. They all remarked on how much fun they had; and in several cases the research team had to stop the participants for a matter of time, as they wanted to keep playing with AR[t]chive. Expressions such as "nice", "cool" and "amazing" were frequently used by participants. Some were eager to start playing around by themselves as soon as they had the headset on.

One of the most appreciated features was the ability to gather objects and attach them to a surface. It produced a "wow" effect and allowed participants to select, organize, and simultaneously inspect those objects. Another associated and well valued interaction was provided through the pin tool, which allows users to establish and document relationships between objects related to different artworks. Most participants used a wall to create their compositions of objects, but some participants also used a table, and even computer screens to attach the objects, creating larger 3D compositions with links across mid-air. One of the participants connected objects on two opposite walls, building parallel lines between the two walls to hang objects (much like a clothesline). One participant stated how these features "would be great to use for their mind maps". Some participants (7) did not use the inventory box to collect objects during the proposed task, because they did not feel the need as things were close enough to be collected and attached to a surface one by one. Other participants found the box useful, especially for interaction in larger physical spaces. These participants enjoyed playing with the box, exploring what happens when they drop it on the floor or when the box becomes full, taking objects out of the box and trying to scroll through contents with a finger.

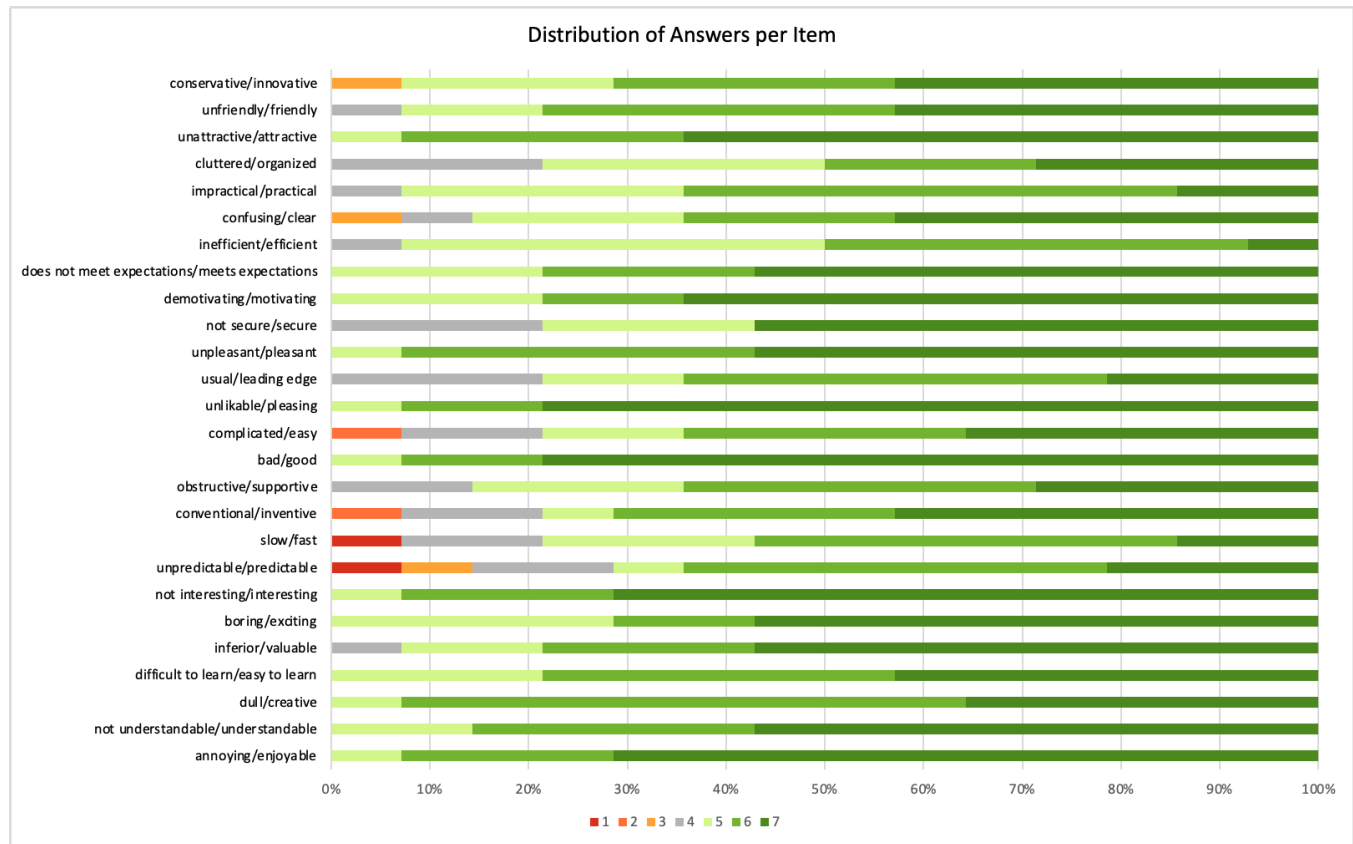


Figure 4: UEQ – Distribution of answers per item

Participants also very much enjoyed playing with the books. They appreciated their visualization as a ribbon of flying pages (Figure 1h), the animated behaviour of the pages, as well as being able to detach individual pages to add to their compositions. They wanted to be able to scale pages to further inspect them and pin books on the walls, which they cannot easily do in the physical world.

Most participants had no problems using the pin tool to link the objects. However, a few of them required a bit of practice at the beginning. Using the hand menu was straightforward for all the participants.

It was interesting to observe how a few participants used both hands at the same time to move objects around, something which could be accomplished with only one hand; and how a few made sure the objects were securely "glued" to the wall before moving on to their next task (they "double-pressed" the object to the wall, much like one may do with real sticky notes). No fatigue using hand gestures was reported or observed during the period of the tests. Although this aspect was not formally assessed, the participants enthusiastically interacted with the system showing no signs of fatigue.

Aspects to improve include the legibility of books' content, as text in the page scans was reportedly difficult to read due to font size, resolution, contrast and also the limited field of view. These

stem from technical limitations related to the device and could conceivably be improved or overcome in due time. Other aspects that can be improved are the ability to manipulate the 3D models of artworks (which is presently very limited, albeit in a conscious way); and having collections (swirls, word clouds) expand further away from the user, instead of forcing the user to step back for a better view. Participants were also engaged enough to offer suggestions for new features, such as the possibility to export the compositions they had created.

5 Conclusion

We have described and discussed early insights into the usability of a digital archive in mixed reality, dubbed AR[t]chive. Results were overall quite positive, with AR[t]chive perceived as having very good usability on a first approach, in free exploration and a lightly structured task. Participants had pleasant and even playful experiences, with little or no friction. The mixed reality archive was perceived as easy and enjoyable to use. Participants appreciated the possibility of attaching virtual objects to surfaces and creating lines between these, employing their preferred strategies and plenty of creativity during study sessions. The ability to examine objects in detail, specifically book scans, was the more pressing issue, which can be addressed in the future.

5.1 Future Work

A larger, more in-depth study would allow us to understand the performance of AR[t]chive concretely as a tool for scholars and researchers, and potentially to a broader audience. This will require a larger number of participants, many who are unfamiliar with the artifact, engaging over longer periods of time, ideally in personal, everyday usage; leading also to a better understanding of the practicality of tools (such as the inventory) or inspiring the development of new ones. It may also imply the porting of AR[t]chive to devices more commonly available (such as the Meta Quest 3). Of great benefit to AR[t]chive in general, but also to a deeper study, would be the inclusion of more artists and works in the archive, for instance with resource to data from ADA.

Acknowledgments

We would like to thank our participants for their time, kindness and valuable input to the study. We also thank Interface Cultures at the University of Arts Linz for accommodating our research. AR[t]chive was developed within the context of the project Lehr- und Forschungsinfrastruktur für Digitale Künste an Hochschulen (LeFo), kindly funded by the Austrian Federal Ministry of Education, Science and Research (BMBWF). This work is also supported by UID/04516/NOVA Laboratory for Computer Science and Informatics (NOVA LINCS) with the financial support of FCT.IP

References

- [1] [n. d.]. *Apple Vision OS 2*. Retrieved January 10, 2025 from <https://www.apple.com/visionos/visionos-2/>
- [2] [n. d.]. *Archive of Digital Art*. Retrieved January 10, 2025 from <https://digitalartarchive.at/>
- [3] [n. d.]. *Steam VR*. Retrieved January 10, 2025 from <https://store.steampowered.com/steamvr>
- [4] 2024. *A New Era for Mixed Reality*. Retrieved January 10, 2025 from <https://www.meta.com/blog/quest/meta-horizon-os-open-hardware-ecosystem-asus-republic-gamers-lenovo-xbox/>
- [5] Ronald T. Azuma. 2017. Making Augmented Reality a Reality, In *Imaging and Applied Optics 2017 (3D, AIO, COSI, IS, MATH, pcAOP)*. *Imaging and Applied Optics 2017 (3D, AIO, COSI, IS, MATH, pcAOP)*, JTu1F.1. <https://doi.org/10.1364/3D.2017.JTu1F.1>
- [6] Daniel Becker. 2016. *Atlas or oracle?* Retrieved January 10, 2025 from <https://zkm.de/en/news/atlas-or-oracle>
- [7] Jill Bennett and iCinema Centre for Interactive Cinema Research. 2008. *T_Visionarium a User's Guide*. Sydney : University of New South Wales Press.
- [8] John Brooke. 2013. SUS: a retrospective. *J. Usability Studies* 8, 2 (Feb. 2013), 29–40.
- [9] J. B. Brooke. 1996. *SUS: A 'Quick and Dirty' Usability Scale*. CRC Press, 207–212. <https://doi.org/10.1201/9781498710411-35>
- [10] Leonardo Candela, Donatella Castelli, and Pasquale Pagano. 2013. Virtual Research Environments: An Overview and a Research Agenda. *Data Science Journal* (Jul 2013). <https://doi.org/10.2481/dsj.GRDI-013>
- [11] Barrett Ens, Juan David Hincapié-Ramos, and Pourang Irani. 2014. Ethereal planes: a design framework for 2D information space in 3D mixed reality environments. In *Proceedings of the 2nd ACM Symposium on Spatial User Interaction* (Honolulu, Hawaii, USA) (*SUI '14*). Association for Computing Machinery, New York, NY, USA, 2–12. <https://doi.org/10.1145/2659766.2659769>
- [12] Philip Grimmer, Emma Krantz, Nagida Helsby-Clark, Dominic Branchaud, and Viveka Weiley. 2023. Dynamic Ocean Explorer: XR Experience. In *SIGGRAPH Asia 2023 XR* (Sydney, NSW, Australia) (*SA '23*). Association for Computing Machinery, New York, NY, USA, Article 8, 1 pages. <https://doi.org/10.1145/3610549.3614616>
- [13] Jens Grubert, Eyal Ofek, Michel Pahud, and Per Ola Kristensson. 2018. The Office of the Future: Virtual, Portable, and Global. *IEEE Computer Graphics and Applications* 38, 6 (2018), 125–133. <https://doi.org/10.1109/MCG.2018.2875609>
- [14] Lucas Joos, Maximilian T. Fischer, Daniel A. Keim, and Johannes Fuchs. 2023. Aesthetic-Driven Navigation for Node-Link Diagrams in VR. In *Proceedings of the 2023 ACM Symposium on Spatial User Interaction* (Sydney, NSW, Australia) (*SUI '23*). Association for Computing Machinery, New York, NY, USA, Article 25, 10 pages. <https://doi.org/10.1145/3607822.3614537>
- [15] UQO Cyberpsychology Lab. 2004. *Revised Presence Questionnaire 3.0*. Retrieved January 10, 2025 from http://w3.uqo.ca/cyberpsy/wp-content/uploads/2019/04/QEP_vf.pdf
- [16] Bettina Laugwitz, Theo Held, and Martin Schrepp. 2008. Construction and Evaluation of a User Experience Questionnaire. In *HCI and Usability for Education and Work*, Andreas Holzinger (Ed.). Springer Berlin Heidelberg, Berlin, Heidelberg, 63–76.
- [17] James R. Lewis and Jeff Sauro. 2018. Item benchmarks for the system usability scale. *J. Usability Studies* 13, 3 (May 2018), 158–167.
- [18] Tiago Martins, Christa Sommerer, and Laurent Mignonneau. 2023. Exploring a digital art archive in mixed reality. *Virtual Creativity* 13, Immersive Horizons: Blurring the Creative Frontiers Between Virtual and Material Worlds (2023), 181–197. https://doi.org/10.1386/vcr_00083_1
- [19] Klaus Schoeffmann, David Ahlström, and Marco A. Hudelist. 2014. 3-D Interfaces to Improve the Performance of Visual Known-Item Search. *IEEE Transactions on Multimedia* 16, 7 (2014), 1942–1951. <https://doi.org/10.1109/TMM.2014.2333666>
- [20] Martin Schrepp. 2023. *User Experience Questionnaire Handbook*. Retrieved January 10, 2025 from <https://www.ueq-online.org/>
- [21] Martin Schrepp, Andreas Hinderks, and Jörg Thomaschewski. 2014. Applying the User Experience Questionnaire (UEQ) in Different Evaluation Scenarios. In *Design, User Experience, and Usability. Theories, Methods, and Tools for Designing the User Experience*, Aaron Marcus (Ed.). Springer International Publishing, Cham, 383–392. https://doi.org/10.1007/978-3-319-07668-3_37
- [22] Christa Sommerer and Laurent Mignonneau. [n. d.]. *Laurent Mignonneau and Christa Sommerer (Artist Website)*. Retrieved January 10, 2025 from <https://www.interface.ug.ac.at/christa-laurent/>
- [23] Bob G. Witmer and Michael J. Singer. 1998. Measuring Presence in Virtual Environments: A Presence Questionnaire. *Presence: Teleoperators and Virtual Environments* 7, 3 (06 1998), 225–240. <https://doi.org/10.1162/105474698565686>